F26-003 FYP-1 Mid-Evaluation

Investify

Educational equity decisions with evidence and checked explanations

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Fall 2026 Report date: 11 October 2026

Problem and research contribution

Make decisions and explanations inspectable.

- Temporal evaluation: use only information available at decision time.
- Evidence: inspect technical/news values and raw-score contributions.
- Checking: reject contradictory visible claims and verify recovery.

News/social divergence and user-comprehension evaluation remain later work.

Scope and acceptance checks

Current scope

- Historical NVDA, five sessions
- Five technical + six news inputs
- Complete capacity/tolerance score
- Low-confidence abstention

Observable acceptance

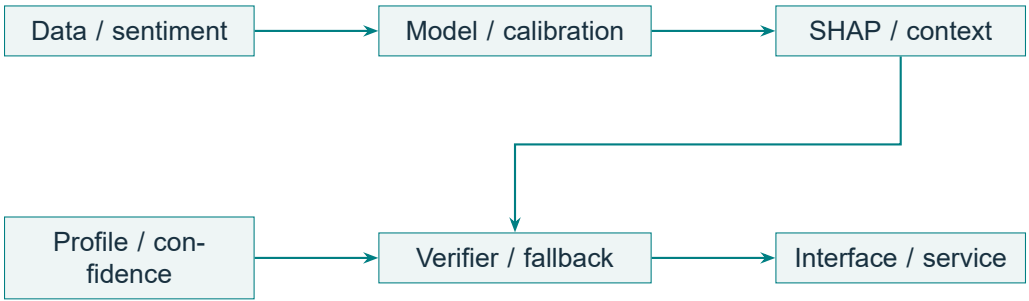
- Validate source/case identity
- Preserve model direction
- Reject wrong prose / JSON
- Check fallback independently

Mid-1 deliverables

Area	Evidence
Report	Detailed use cases, grouped requirements, metrics and individual tests
Design	Six-module architecture, keys/cardinalities and behavioral models
Live core	Evidence, eighteen-answer profile, uncertainty and checking/recovery
Process	Member delivery, dated plan, Windows/Ubuntu CI and verified artifacts

Supervisor checkpoint; planned presentation/demo duration: fifteen minutes.

System architecture



Model probabilities remain fixed during profile calculation and explanation checking.

Domain model and file schema

- Experiment identity: source + artifact digest
- Case identity: experiment + session
- Record: class, probabilities, feature/SHAP maps and model hashes
- Profile/candidate/result: local state with current identity binding

The report supplies attributes, keys and cardinalities. Storage uses file/session contracts; no SQL database is claimed.

Explanation check and recovery

1. Read the current record and visible candidate.
2. Verify prose and structured claims independently.
3. Retain candidate rejection; construct deterministic fallback.
4. Verify fallback, then show current output or an explicit error.

Correct JSON cannot conceal an incorrect displayed probability.

Live demonstration

1. **Decision**: diagnostic class/probabilities and uncertainty
2. **Evidence**: technical and news feature meaning/effect
3. **Risk**: published tolerance score, allowance and confidence requirement
4. **Explanation**: inject 1.00%, reject and verify fallback

Multiclass frozen NVDA / 21 December 2023. Calculations/checking run live; model outputs are recorded.

Calibration correction and abstention

Frozen mode	Buy/Hold/Sell	Macro-F1
Raw model	40 / 51 / 95	0.3182
Independent isotonic	186 / 0 / 0	0.2238
Global temperature	40 / 51 / 95	0.3182

The correction preserves order, not useful signal.

Temperature reaches its upper bound; probabilities approach one-third. The interface withholds directional suggestions.

Temporal leakage experiment

Five-session arm	Macro-F1	Lift
Clean	0.2843	-0.0444
Shuffled	0.4457	+0.1142
Centred Bollinger	0.5390	+0.2104

No-purge and same-bar variants do not guarantee inflation. Endpoint purge and the configured calendar-day gap are distinct controls.

Sentiment and teacher studies

Social comparison

Same 24 held-out references

Candidate	Macro-F1
CryptoBERT	0.2564
Cardiff	0.6016

Crypto probes return positive/negative; equity outputs remain all neutral.

Teacher construction

- Actual recorded prose
- Deterministic factual targets
- Student remains untrained
- Larger comparison is later

Small reference sample; independent validation and deployment selection remain open.

Delivered work and next milestones

Member	Delivered part / continuation
Ahmed	Temporal data, leakage and CI/delivery; parity/deployment next
Rafay	Interface, profile and sentiment; divergence/acceptance next
Soban	Engine, calibration, SHAP and verifier; student/QLoRA next

One-third responsibility and five minutes each. Internal dated targets are in the report; current code practice continues.